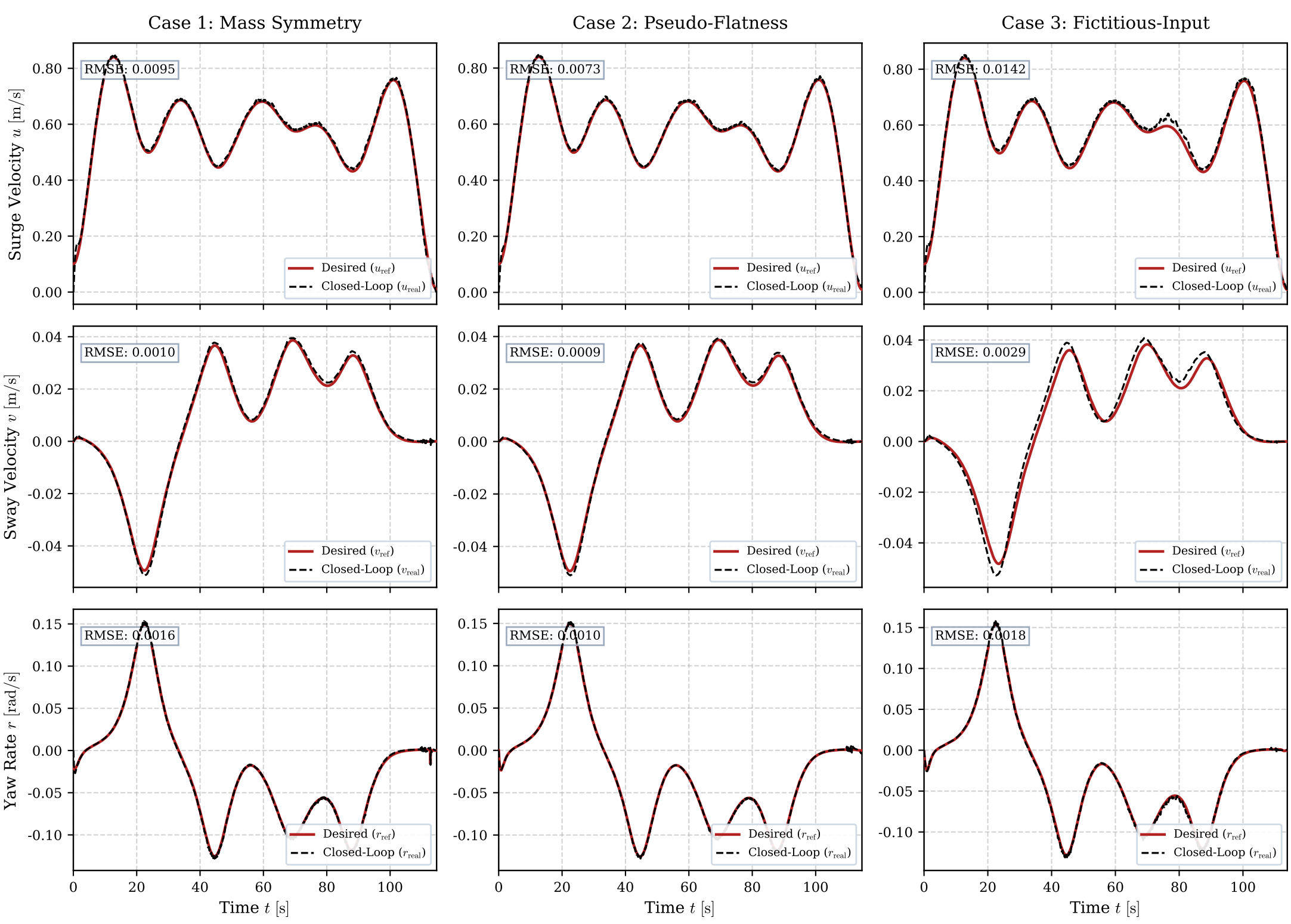

RealtimeController Velocity Tracking Performance (u, v, r Comparison)
Case 1: Mass Symmetry
Case 2: Pseudo-Flatness
Case 3: Fictitious-Input
Surge Velocity u [m/s]
Sway Velocity v [m/s]
Yaw Rate r [rad/s]
Time t [s]
RMSE: 0.0095
RMSE: 0.0073
RMSE: 0.0142
RMSE: 0.0010
RMSE: 0.0009
RMSE: 0.0029
RMSE: 0.0016
RMSE: 0.0010
RMSE: 0.0018
Desired (u_{ref})
Closed-Loop (u_{real})
Desired (v_{ref})
Closed-Loop (v_{real})
Desired (r_{ref})
Closed-Loop (r_{real})